

Analysis of $f(x, y) = \sqrt{|xy|}$ at the Origin

Problem Statement

Prove that the function,

$$f(x, y) = \sqrt{|xy|}.$$

is not differentiable at the point $(0,0)$, but that f_x and f_y both exist at the origin and have the value 0. Hence deduce that these two partial derivatives are continuous except at the origin.

Solution

Study the following at $(0,0)$:

1. Existence of the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.
2. Differentiability of f at $(0,0)$.
3. Continuity of the partial derivatives.

Step 1: Partial Derivatives at $(0,0)$

Using the limit definition,

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{|h \cdot 0|}}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

By symmetry,

$$f_y(0,0) = 0.$$

Step 2: Differentiability at $(0,0)$

If f were differentiable at $(0,0)$, then since $f(0,0) = 0$ and both partial derivatives vanish there, the following limit would equal zero:

$$L = \lim_{(h,k) \rightarrow (0,0)} \frac{f(h,k) - f(0,0) - f_x(0,0)h - f_y(0,0)k}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{\sqrt{|hk|}}{\sqrt{h^2 + k^2}}.$$

Take the path $k = mh$ ($m \neq 0$, $h \rightarrow 0$):

$$\frac{\sqrt{|h \cdot mh|}}{\sqrt{h^2 + m^2h^2}} = \frac{|h|\sqrt{|m|}}{|h|\sqrt{1+m^2}} = \frac{\sqrt{|m|}}{\sqrt{1+m^2}}.$$

For $m = 1$ this gives $\frac{1}{\sqrt{2}} \neq 0$.

Since the limit depends on the path, $L \neq 0$, and therefore f is **not differentiable** at $(0,0)$.

Step 3: Partial Derivatives Away from the Axes

For $xy \neq 0$, the chain rule gives

$$f_x(x, y) = \frac{1}{2}|xy|^{-1/2} \cdot \frac{\partial}{\partial x}|xy| = \frac{y \operatorname{sgn}(xy)}{2\sqrt{|xy|}},$$

or equivalently,

$$f_x(x, y) = \frac{\operatorname{sgn}(x)\sqrt{|y|}}{2\sqrt{|x|}}.$$

Similarly,

$$f_y(x, y) = \frac{x \operatorname{sgn}(xy)}{2\sqrt{|xy|}} = \frac{\operatorname{sgn}(y)\sqrt{|x|}}{2\sqrt{|y|}}.$$

Step 4: Continuity of the Partial Derivatives

Non-continuity at $(0, 0)$. Consider f_x as $(x, y) \rightarrow (0, 0)$ along two paths.

- Along $y = x^2$ ($x > 0$):

$$f_x(x, x^2) = \frac{\sqrt{x^2}}{2\sqrt{x}} = \frac{x}{2\sqrt{x}} = \frac{\sqrt{x}}{2} \rightarrow 0.$$

- Along $y = x$ ($x > 0$):

$$f_x(x, x) = \frac{\sqrt{x}}{2\sqrt{x}} = \frac{1}{2}.$$

The two path limits differ, so $\lim_{(x,y) \rightarrow (0,0)} f_x(x, y)$ does not exist. Hence f_x is **not continuous at $(0, 0)$** .

Non-existence on the coordinate axes. Consider the point $(0, 1)$. By definition,

$$f_x(0, 1) = \lim_{h \rightarrow 0} \frac{f(h, 1) - f(0, 1)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{|h|}}{h}.$$

For $h > 0$ this equals $h^{-1/2} \rightarrow +\infty$, so the limit does not exist.

In general, $f_x(0, y)$ does not exist for any $y \neq 0$, and by symmetry, $f_y(x, 0)$ does not exist for any $x \neq 0$.

Summary.

Point	f_x exists?	f_y exists?
$xy \neq 0$	Yes	Yes
$(0, 0)$	Yes ($= 0$)	Yes ($= 0$)
$x = 0, y \neq 0$	No	Yes
$y = 0, x \neq 0$	Yes	No

Conclusion. The partial derivatives f_x and f_y :

- exist and are continuous on the open region $\{(x, y) : xy \neq 0\}$,
- exist (and equal zero) at $(0, 0)$, but are **not continuous** there,
- **do not exist** on most of the coordinate axes (excluding the origin).

Hence the claim that “the partial derivatives are continuous everywhere except at $(0, 0)$ ” is **incorrect**; they fail to exist on the entire y -axis (excluding the origin) for f_x , and on the entire x -axis (excluding the origin) for f_y .